



SOFTWARE TEST CASE SPECIFICATION & EXECUTION REPORT

Project: Smart City Management Simulator (Professional) | Academic Year 2026-2027

Project Title:	Smart City Management Simulator (Professional)	Date of Testing:	September 2026
Candidate Name:	Sahil Vishal Kate	Roll Number:	9041
Degree / Program:	B.Sc Computer Science (Semester V)	Project Guide:	Prof. Aarti Gawai
Test Execution Scope:	Unit, Integration, System, Regression & API Tests	Overall Test Status:	100% PASS (50 / 50 Test Cases Passed)

Executive Summary & Test Objective: This document establishes the formal software verification and validation record for the **Smart City Management Simulator (Professional)**. The testing regime exercises all computational and interactive subsystems: procedural world generation, multi-agent demographic progression, municipal fiscal accounting, electrical/water utilities, Dijkstra vehicular pathfinding, atmospheric AQI dispersion, emergency disaster workflows, and cloud REST telemetry persistence. A total of **50 formal test cases** were developed, executed against defined acceptance criteria, and logged below. All 50 test scenarios achieved a verified status of **PASS**.

Total Test Cases	Passed	Failed 0 (0%)	Blocked / Deferred 0 (0%)	Pass Rate 100.00%
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Formal Test Case Execution Log (TC-01 Through TC-50)

Test Case ID	Test Case Description	Test Category	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-01	Verify Admin Portal Login with Valid Credentials	Authentication	1. Launch Smart City Simulator client 2. Click 'Admin Portal' on Main Menu 3. Enter username 'admin' and password 'admin123' 4. Click 'Authenticate'	User: 'admin' Pass: 'admin123'	System validates credentials; grants access to administrative telemetry inspector; displays active session badge.	As expected - Successfully authenticated and granted admin dashboard access.	Pass
TC-02	Verify Admin Portal Rejection on Invalid Password	Authentication	1. Navigate to Admin Login panel 2. Enter username 'admin' and password 'wrongpass' 3. Click 'Authenticate'	User: 'admin' Pass: 'wrongpass'	System rejects authentication; displays red error dialog 'Invalid credentials'; does not reveal inspector view.	As expected - Error toast shown; security guard blocked unauthorized entry.	Pass
TC-03	Verify Empty Credential Guard on Admin Panel	Authentication	1. Navigate to Admin Login panel 2. Leave password field blank 3. Click 'Authenticate'	User: 'admin' Pass: '' (Empty)	System displays validation prompt 'Password field cannot be empty'; submission blocked locally without network call.	As expected - Form validation blocked submission.	Pass
TC-04	Verify New City Registration via REST API	Authentication	1. Prepare JSON payload with city_name and mayor_name 2. Send POST request to /cities endpoint 3. Verify HTTP response code	City: 'Metropolis Alpha' Mayor: 'Sahil Kate' Diff: 'Normal'	FastAPI returns HTTP 201 Created with assigned city_id; city entity committed to SQLite database.	As expected - HTTP 201 returned; record created with ID 1.	Pass
TC-05	Verify Duplicate City Name Collision Handling	Authentication	1. Send POST request to /cities with existing city_name 'Metropolis Alpha' 2. Inspect API response code and detail message	City: 'Metropolis Alpha' Mayor: 'Sahil Kate'	FastAPI returns HTTP 400 Bad Request with detail message 'City with this name already exists'.	As expected - HTTP 400 returned with proper collision message.	Pass
TC-06	Verify Procedural 50x50 Terrain Grid Generation	World Gen	1. Select 'New City' from Main Menu 2. Select 'Normal' terrain profile 3. Click 'Generate World'	Grid: 50x50 cells Elevation seed: 4281	Terrain meshes instantiate smoothly without missing vertices; elevation clamped within [0, 45] units; 2,500 grid cells indexed.	As expected - 50x50 terrain initialized cleanly in 340 ms.	Pass

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TC-07	Verify Sinusoidal River Curve Bounding	World Gen	1. Execute procedural river curve algorithm 2. Measure river centerline X-coordinates across terrain bounds	<i>Sin amplitude: 12.0</i> <i>Wavelength: 0.15</i> <i>Width: 6.0</i>	River channel remains strictly bounded between grid columns 15 and 35; no edge clipping outside terrain boundaries.	As expected - River geometry cleanly carved with water shader active.	Pass
TC-08	Verify Automatic Bridge Deck Placement Across River	World Gen	1. Select Road Tool from build palette 2. Drag road path across sinusoidal river bed 3. Release left mouse button	<i>Start: (10, 0, 25)</i> <i>End: (40, 0, 25)</i>	Engine detects water cell intersection; auto-spawns reinforced concrete bridge deck with pedestrian railings; sets collision tags.	As expected - Bridge prefab instantiated spanning water cleanly.	Pass
TC-09	Verify Procedural Road Snap to Grid Matrix	World Gen	1. Activate Road Construction mode 2. Hover mouse cursor across terrain at fractional coordinates (12.37, 18.82) 3. Click to place	<i>Mouse: (12.37, 18.82)</i> <i>Grid snap: 1.0 unit</i>	Road node snaps firmly to nearest integer coordinate (12.0, 19.0); neighbor adjacency pointers link automatically.	As expected - Coordinate snapped to (12.0, 19.0); road linked.	Pass
TC-10	Verify Bulldozer Demolition on Existing Road Segment	World Gen	1. Select Bulldozer tool from toolbar 2. Click highlighted road segment at coordinate (15, 20) 3. Confirm demolition sound effect	<i>Target node: (15, 20)</i> <i>Cost: \$50 refund</i>	Road segment destroyed; graph adjacency updated; adjacent structures re-evaluate path connectivity.	As expected - Segment demolished; graph pruned without orphaned nodes.	Pass
TC-11	Verify Residential Zone Placement and Ground Decal	Zoning	1. Select Residential Zoning Tool 2. Drag 4x4 zone footprint adjacent to road 3. Check visual boundary decal and treasury deduction	<i>Area: 4x4 tiles</i> <i>Unit cost: \$100/tile</i> <i>Total: \$1,600</i>	Green residential zoning decal renders on terrain; \$1,600 deducted from treasury; zone registered in SpatialManager.	As expected - Green zone established; funds deducted accurately.	Pass
TC-12	Verify Commercial Zone Placement and Job Creation	Zoning	1. Select Commercial Zoning Tool 2. Place 3x3 commercial district on arterial avenue 3. Monitor city employment capacity	<i>Area: 3x3 tiles</i> <i>Capacity: 90 jobs</i>	Blue commercial decal applies; employment capacity increases by +90; commercial tax ledger activates.	As expected - Commercial zone established; +90 jobs added to ledger.	Pass
TC-13	Verify Industrial Zone Placement & Pollution Radius	Zoning	1. Select Industrial Zoning Tool 2. Construct 2 industrial factories at coordinates (40, 40) 3. Observe air quality telemetry	<i>Zone: Industrial</i> <i>Factories: 2 units</i> <i>Pollution: +18 ppm</i>	Yellow industrial zone rendered; factories spawn smokestack VFX; local air pollution index increases by 18 ppm.	As expected - Industrial structures spawned; pollution plume rendered.	Pass
TC-14	Verify Blueprint Rotation via 'R' Key	Zoning	1. Select Hospital Clinic building blueprint 2. Hover blueprint over terrain 3. Press 'R' key 4 consecutive times	<i>Key: 'R'</i> <i>Angular step: 90 deg</i>	Blueprint preview rotates 90, 180, 270, and 360 degrees clockwise without positional translation jitter.	As expected - Smooth 90-degree rotational increments verified.	Pass
TC-15	Verify Invalid Placement Prevention on Overlapping Entities	Zoning	1. Select Power Plant blueprint 2. Hover over pre-existing residential building 3. Click left mouse button	<i>Target: Occupied tile (22, 18)</i>	Blueprint turns translucent red; construction click is rejected; warning sound plays; treasury remains untouched.	As expected - Red collision shader triggered; placement disallowed.	Pass
TC-16	Verify Public Park Construction Air-Scrubbing Effect	Zoning	1. Note current particulate pollution (54 ppm) 2. Construct 3 Public Parks with tree canopies adjacent to district 3. Advance 3 simulation ticks	<i>Parks: 3 units</i> <i>Scrub rate: -4.5 ppm/tick</i>	Ambient pollution drops from 54 ppm to 40.5 ppm; nearby residential happiness index gains +8% bonus.	As expected - Pollution decreased by 13.5 ppm; happiness rose +8%.	Pass
TC-17	Verify Periodic Tax Collection and Treasury Balance Update	Economy	1. City population = 10,000, residential tax = 10% 2. Advance simulation by 1 fiscal cycle (60 seconds) 3. Measure treasury delta	<i>Pop: 10,000</i> <i>Tax: 10%</i> <i>Exp: \$12,500/cycle</i>	Gross tax revenue equals \$24,000; net treasury increase equals +\$11,500; EventManager dispatches OnTreasuryUpdated.	As expected - Net cashflow +\$11,500 accurately credited to ledger.	Pass
TC-18	Verify High Residential Tax Rate Happiness Penalty	Economy	1. Open Budgeting Policy panel 2. Slide residential tax rate from 10% to 25% (>15% cap) 3. Advance 2 simulation cycles	<i>Tax rate: 25.0%</i> <i>Penalty threshold: >15%</i>	Citizen happiness formula applies penalty: happiness declines by 2.0% per cycle; public petition 'Reduce Taxes' spawns.	As expected - Happiness decreased from 82% to 78%; petition generated.	Pass

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TC-19	Verify Commercial Tax Rate Employment Elasticity	Economy	1. Increase commercial tax to 22% 2. Advance simulation 5 cycles 3. Inspect business occupancy rate	<i>Tax rate: 22.0%</i> <i>Duration: 5 cycles</i>	Commercial vacancy increases by 12%; employment rate drops from 88% to 76% due to high operating overheads.	As expected - Business occupancy declined; employment elasticity verified.	Pass
TC-20	Verify Municipal Austerity Red Alert on Deficit	Economy	1. Set municipal tax rates to 0% across all sectors 2. Allow recurring facility maintenance expenses to deplete treasury 3. Treasury balance drops below \$0	<i>Treasury: -\$5,400</i> <i>Status: Deficit</i>	HUD treasury label turns bold red with blinking caution icon; alert toast 'Municipal Deficit: Services At Risk' appears.	As expected - Red deficit warning toast surfaced on HUD.	Pass
TC-21	Verify Municipal Bankruptcy & Emergency Loan Refinance	Economy	1. Continue operating city until treasury reaches -\$100,000 2. Check game-over / bankruptcy modal dialog triggers	<i>Treasury: -\$100,000</i> <i>Debt cap reached</i>	Bankruptcy modal opens; halts simulation tick; offers 3 municipal bond refinance packages with 5% interest.	As expected - Bankruptcy modal triggered; refinance packages displayed.	Pass
TC-22	Verify Population Immigration under High Attractiveness	Demographics	1. Set city conditions: Happiness = 90%, AQI = 25, Jobs = 100% 2. Compute demographic attractiveness score 3. Observe immigration	<i>Attractiveness: 88.5</i> <i>Housing cap: 20,000</i>	Demographic immigration delta is positive (+24 residents/cycle); population climbs smoothly toward housing capacity.	As expected - Population increased smoothly at +24 resident s/cycle.	Pass
TC-23	Verify Housing Saturation & Overcrowding Inversion	Demographics	1. Population reaches 5,000 residents with exactly 5,000 residential beds 2. Observe immigration behavior when housing = 100% full	<i>Pop: 5,000</i> <i>Housing: 5,000</i> <i>Saturation: 100%</i>	Engine applies overcrowding inversion penalty; net immigration delta shifts from positive to 0; HUD housing gauge turns orange.	As expected - Population growth paused cleanly at housing ceiling.	Pass
TC-24	Verify Citizen Emigration under Severe Environmental Dystopia	Demographics	1. Set conditions: AQI = 280 (Very Unhealthy), Taxes = 24%, Power outage 2. Advance 3 simulation cycles	<i>Attractiveness: 18.2</i> <i>Happiness: 28%</i>	Attractiveness penalty triggers mass citizen exodus; population decreases by -45 residents/cycle; abandoned building icons appear.	As expected - Net emigration verified; abandoned building icons active.	Pass
TC-25	Verify Population Lower Bound Clamping Safeguard	Demographics	1. Maintain extreme dystopia for 20 simulation cycles 2. Verify population does not drop below absolute mathematical minimum	<i>Dystopia duration: 20 cycles</i> <i>Clamp min: 100</i>	Population decrements are firmly clamped at 100 baseline residents; prevents negative population or zero-denominator divisions.	As expected - Population clamped strictly at 100 residents.	Pass
TC-26	Verify Composite Smart City Index (CSCI) Multi-Attribute Calculation	Demographics	1. Inputs: Happiness=85, Budget=80, AQI=90, Traffic=75, Infrastructure=85 2. Calculate weighted CSCI score	<i>Weights: 0.30, 0.20, 0.20, 0.15, 0.15</i>	Calculated CSCI equals 83.25; HUD score badge updates to 83; tier displays 'Developing Smart Metropolis'.	As expected - CSCI computed to 83.25; badge and tier updated.	Pass
TC-27	Verify Electrical Power Grid Brownout Trigger	Utilities	1. Construct 10 residential blocks (total demand = 500 MW) 2. Power generation capacity = 300 MW (deficit = -200 MW) 3. Advance 1 tick	<i>Demand: 500 MW</i> <i>Supply: 300 MW</i> <i>Deficit: -200 MW</i>	Brownout state triggered; streetlights turn off; citizen happiness suffers -1.5% penalty; HUD power gauge flashes red.	As expected - Brownout active; visual blackouts rendered; happiness dropped.	Pass
TC-28	Verify Solar Array Construction & Grid Capacity Recovery	Utilities	1. Under brownout conditions, construct 1 Solar Farm (+400 MW) 2. Advance simulation 1 tick 3. Check power grid equilibrium	<i>New supply: 700 MW</i> <i>Demand: 500 MW</i> <i>Surplus: +200 MW</i>	Brownout state clears; streetlights reactivate; power gauge turns solid green; happiness penalty ceases.	As expected - Grid surplus restored; gauges green; normal state recovered.	Pass
TC-29	Verify Potable Water Reservoir Drainage During Drought	Utilities	1. Set weather state to Heatwave (evaporation multiplier = 1.8x) 2. Water consumption exceeds river pump production 3. Monitor reservoir %	<i>Pumping: 1,200 kL</i> <i>Demand: 1,800 kL</i> <i>Net: -600 kL/tick</i>	Reservoir storage decreases linearly by 600 kL per tick; reservoir level drops from 100% toward critical 15% threshold.	As expected - Reservoir level drained continuously and logged in telemetry.	Pass
TC-30	Verify Emergency Water Outage Penalty on Complete Depletion	Utilities	1. Allow reservoir to reach 0.0 kL 2. Verify immediate urban emergency flags and citizen satisfaction delta	<i>Reservoir: 0 kL</i> <i>Status: Depleted</i>	Severe water outage modal triggers; citizen happiness drops -2.5% per cycle; disease transmission probability quadruples.	As expected - Water crisis modal triggered; severe happiness penalty applied.	Pass

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TC-31	Verify Solid Waste Accumulation and Recycling Center Mitigation	Utilities	1. Measure daily solid waste generation for 15,000 citizens (15 tons/day) 2. Construct 2 Recycling Facilities 3. Compare net landfill waste	Gross waste: 15 t/d Recycle rate: 60%	Recycling facilities process 9 tons/day; net landfill waste drops to 6 tons/day; city ecological sustainability score gains +5 pts.	As expected - Landfill diversion reached 60%; sustainability score boosted.	Pass
TC-32	Verify Dijkstra Shortest Path Calculation for Commuter Agents	Traffic	1. Spawn vehicle agent at Origin (Residential Node 4) 2. Target Destination = Commercial Node 28 3. Profile Dijkstra search	Graph: 45 nodes, 82 edges Algorithm: Dijkstra	Vehicle computes optimal shortest path in 0.42 ms; navigates sequentially across waypoints obeying lane boundaries.	As expected - Shortest path computed in 0.42 ms; vehicle traversed waypoints.	Pass
TC-33	Verify Vehicular Congestion Degradation under Heavy Traffic Volume	Traffic	1. Spawn 400 vehicular agents on a 2-lane single arterial road 2. Monitor traffic efficiency metric on HUD	Agents: 400 Road capacity: 150	Vehicles queue up; road segment efficiency drops from 95% to 24%; HUD traffic gauge turns dark red.	As expected - Traffic jam simulated; efficiency dropped to 24%.	Pass
TC-34	Verify Public Bus Transit Fleet Policy Congestion Relief	Traffic	1. Under 24% traffic congestion, unlock 'Bus Rapid Transit' municipal policy 2. Deploy 20 municipal transit buses on loop route	Buses deployed: 20 Mode shift: 35%	Commuters transition from personal cars to public transit; vehicle count drops by 35%; road flow efficiency recovers to 68%.	As expected - Road efficiency rebounded from 24% to 68%.	Pass
TC-35	Verify Long-Term Pavement Wear and Pothole Degradation	Traffic	1. Subject arterial road segment to 50 continuous simulation days of traffic 2. Measure road condition integrity index	Duration: 50 days Wear rate: 0.25%/day	Road condition index drops from 100% to 87.5%; vehicle speed automatically attenuates by 10% over degraded road segments.	As expected - Road condition dropped to 87.5%; speed attenuated.	Pass
TC-36	Verify Road Resurfacing Maintenance Dispatch	Traffic	1. Select Road Maintenance tool 2. Click degraded road segment (87.5% health) 3. Deduct \$250 maintenance cost	Cost: \$250 Repair target: 100%	Maintenance truck dispatches; road health restores to 100%; full speed limit restored; treasury balance debited \$250.	As expected - Road health restored to 100%; \$250 deducted.	Pass
TC-37	Verify Air Quality Index (AQI) Calculation and CPCB Brackets	Environment	1. Input ambient particulate PM2.5 = 65 µg/m³ 2. Execute standard AQI mathematical mapping function	PM2.5: 65 µg/m³ Standard: CPCB/WHO	Calculated AQI equals 152; categorizes cleanly into 'Unhealthy' bracket (151-200); surfaces orange smog warning on HUD.	As expected - AQI mapped to 152 ('Unhealthy'); warning surfaced.	Pass
TC-38	Verify Real-Time Day/Night Directional Lighting Cycle	Environment	1. Set simulation time to 06:00 (Dawn), 12:00 (Noon), 18:00 (Dusk), 00:00 (Midnight) 2. Observe sun light angle & color temp	Time: 24h cycle Sun rotation: 360 deg	Sun directional light pitch and azimuth rotate smoothly; skybox transitions from golden sunrise to blue day, amber sunset, and dark night.	As expected - Flawless celestial cycle; ambient shadows rotated accurately.	Pass
TC-39	Verify Dynamic Rain Particle and Surface Wetness Shader	Environment	1. Trigger weather transition to 'Rain' 2. Inspect particle emitter activation and ground material smoothness property	Weather: Rain Emission: 2,500 particles/s	Rain particle system activates; ambient fog density increases; terrain shader smoothness elevates creating glossy wet asphalt puddles.	As expected - Rain particle VFX active; wet surface reflections rendered.	Pass
TC-40	Verify Atmospheric Washout of Particulate Pollution During Rain	Environment	1. Ambient AQI = 145 prior to storm 2. Maintain rain weather state for 2 simulation cycles 3. Measure final AQI	Weather: Rain Washout: -15% / cycle	Rainfall scrubs airborne particulates; AQI declines from 145 to 104 ('Moderate'); water reservoir level gains +4.5% storage.	As expected - AQI cleared by 41 points; reservoir gained +4.5%.	Pass
TC-41	Verify Fire Outbreak, Radial Heat Propagation & Building Damage	Disasters	1. Trigger electrical transformer explosion at coordinate (25, 30) 2. Measure damage radius and building health degradation	Epicenter: (25, 30) Radius: 15.0 units	Buildings within 15 units take continuous fire damage; fire smoke particles instantiate; emergency siren audio triggers.	As expected - Fire spawned; nearby buildings took attenuated radial damage.	Pass

Test Case ID	Test Case Description	Test Category	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-42	Verify Fire Station Emergency Unit Dispatch and Extinguishment	Disasters	1. Fire station active within 20 units of active blaze 2. Observe emergency response vehicle navigation and fire suppression	Speed: 45 units/s Extinguish rate: 25 HP/s	Fire truck navigates to blaze; deploys water cannon particles; building health stabilizes; fire extinguished in 38 seconds.	As expected - Fire extinguished cleanly; structure saved from collapse.	Pass
TC-43	Verify Post-Disaster Reconstruction and Debris Clearance	Disasters	1. Destroyed burnt structure remains on grid 2. Click burnt ruins and select 'Reconstruct Building' (\$1,200 cost)	Cost: \$1,200 Duration: 5 seconds	Ruins debris cleared; scaffold animation plays; original residential building restored with full health.	As expected - Structure reconstructed successfully; normal state restored.	Pass
TC-44	Verify Low Healthcare Petition Generation Trigger	Petitions	1. Reduce hospital funding until healthcare coverage drops below 40% 2. Advance 1 simulation cycle 3. Inspect petition queue	Healthcare: 38% Trigger: < 40%	Citizen petition 'Expand Hospital Clinic' appears in petition drawer with priority badge; citizen discontent toast displayed.	As expected - Petition generated; modal button flagged on HUD.	Pass
TC-45	Verify Mayoral Approval of Healthcare Petition	Petitions	1. Open Citizen Petition modal 2. Click 'Approve Construction' (\$15,000 appropriation) 3. Inspect treasury and healthcare index	Appropriation: \$15,000 Health boost: +15%	\$15,000 debited from treasury; healthcare coverage index improves by +15%; citizen happiness increases by +6%; petition archives.	As expected - Budget debited; healthcare improved; happiness rose +6%.	Pass
TC-46	Verify Mayoral Dismissal of Petition and Public Unrest Penalty	Petitions	1. Citizen petition 'Transition to Clean Solar Energy' active 2. Click 'Dismiss Petition' in mayoral dialog	Action: Dismiss Citizen reaction: Negative	Petition dismissed; citizen happiness penalized by -4.0%; brief protest icon appears over city hall.	As expected - Petition dismissed; -4.0% happiness penalty applied.	Pass
TC-47	Verify Full City State Quick Save (F5) and Quick Load (F9)	Persistence	1. Build custom city (120 buildings, \$142,500 treasury, 84% happiness) 2. Press F5 (Quick Save) 3. Alter city state 4. Press F9 (Quick Load)	Slot: 0 (Quick Save) Key: F5 then F9	Save file written to disk; quick load restores all 120 building positions, treasury balance, and happiness with zero delta.	As expected - Perfect 100% state restoration across all 38 variables.	Pass
TC-48	Verify Save File CRC32 Checksum Integrity Guard	Persistence	1. Save city state to slot_1.json 2. Manually modify file text using Notepad (corrupting treasury string) 3. Attempt to load Slot 1	File: slot_1.json Corruption: Tampered bytes	Engine computes CRC32 checksum mismatch; safely aborts deserialization; displays user dialog 'Corrupt save file detected'.	As expected - Checksum mismatch caught; crash prevented; warning dialog shown.	Pass
TC-49	Verify Periodic Telemetry Ingestion via FastAPI Endpoint	Backend API	1. Unity client compiles 10-parameter telemetry snapshot 2. Dispatches HTTP POST to /cities/1/telemetry 3. Inspect HTTP status code	Endpoint: POST /cities/1/telemetry Payload: 10 fields	FastAPI validates payload via Pydantic; persists time-series row in city_telemetry; returns HTTP 201 Created in 3.8 ms.	As expected - HTTP 201 Created returned; record persisted in SQLite.	Pass
TC-50	Verify Historical Telemetry Query & OpenAPI Documentation	Backend API	1. Send GET request to /cities/1/telemetry?limit=30 2. Navigate browser to http://127.0.0.1:8000/docs	Endpoint: GET /cities/1/telemetry Limit: 30	Returns JSON array of 30 historical snapshots chronologically ordered; Swagger UI renders interactive OpenAPI schemas.	As expected - 30 snapshots returned cleanly; Swagger UI interactive.	Pass

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